An acyclic edge coloring of K_8 with no acyclic completion

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Abstract

Alon, Sudakov, and Zaks suggested that every acyclic edge coloring of K_{2n} with $2n+1$ colors might extend to an acyclic edge coloring of K_{2n+1} without introducing new colors. We give an explicit counterexample on K_8. The coloring has exactly two proper extensions to K_9 with the same palette, and each contains a cycle using only colors 2 and 4. We provide certificates for all two-color subgraphs, so the proof can be checked directly from the displayed matrix.

1 Introduction

An edge coloring of a finite simple graph is *acyclic* if it is proper and the union of every two color classes is a forest. In Section 1.2, printed page 160, Alon, Sudakov, and Zaks [1] suggested that every acyclic edge coloring of K_{2n} with $2n+1$ colors might be completed to an acyclic edge coloring of K_{2n+1} without introducing new colors. Here completion preserves the colors of all existing edges.

We give an explicit coloring for which such a completion is impossible.

Theorem 1. *There is a proper acyclic edge coloring of K_8 with palette $\{1, \dots, 9\}$ that has no proper acyclic extension to K_9 with the same palette.*

The construction and the nonextension statement have also been formalized in Lean.[1]

[1] The source files are available at https://github.com/alexgocardinal/acyclic-completion. The theorem `AcyclicCompletion.certified_counterexample` proves properness, acyclicity, and nonextension, using the project's explicit encodings of edge colorings and finite cycles. The formalization uses Lean's standard library.

2 The coloring

Let the vertices of K_8 be $0, \ldots, 7$. For distinct u, v, define $c(uv)$ by the following symmetric matrix; the diagonal entries do not represent edges.

	0	1	2	3	4	5	6	7
0	–	7	8	6	4	3	1	5
1	7	–	6	1	8	9	5	4
2	8	6	–	9	1	5	2	7
3	6	1	9	–	7	8	3	2
4	4	8	1	7	–	2	9	3
5	3	9	5	8	2	–	4	1
6	1	5	2	3	9	4	–	6
7	5	4	7	2	3	1	6	–

Lemma 1. *The coloring c is proper and acyclic.*

Proof. The seven entries off the diagonal in each row are distinct, so c is proper. Table 1 lists the path components of every two-color subgraph, including isolated vertices. For each pair, the listed paths have disjoint vertex sets covering all eight vertices, and their consecutive edges are exactly the edges carrying either of the two colors in the matrix. Thus each two-color subgraph is a disjoint union of paths and is a forest. □

Colors	Path components	Colors	Path components
$1,2$	`06245731`	$3,7$	`105; 27436`
$1,3$	`13605742`	$3,8$	`147; 20536`
$1,4$	`24065713`	$3,9$	`051; 23647`
$1,5$	`31607524`	$4,5$	`2561704; 3`
$1,6$	`42130675`	$4,6$	`21765; 304`
$1,7$	`57243106`	$4,7$	`271043; 56`
$1,8$	`60241357`	$4,8$	`20417; 356`
$1,9$	`06423157`	$4,9$	`046517; 23`
$2,3$	`0547362; 1`	$5,6$	`3076125; 4`
$2,4$	`04562; 173`	$5,7$	`34; 527016`
$2,5$	`073; 16254`	$5,8$	`35207; 416`
$2,6$	`037621; 45`	$5,9$	`07; 325164`
$2,7$	`01; 543726`	$6,7$	`4301276; 5`
$2,8$	`026; 14537`	$6,8$	`412035; 67`
$2,9$	`0; 1546237`	$6,9$	`03215; 467`
$3,4$	`1740563; 2`	$7,8$	`5341027; 6`
$3,5$	`163; 25074`	$7,9$	`015; 64327`
$3,6$	`12; 476305`	$8,9$	`0235146; 7`

Table 1: Certificates for the two-color subgraphs. Each word lists the vertices of a path in order; a semicolon separates components. A one-letter word denotes an isolated vertex.

3 Nonextension

For a vertex v, let $M(v)$ be the set of colors absent from its incident edges. The matrix gives

v	0	1	2	3	4	5	6	7
$M(v)$	$\{2,9\}$	$\{2,3\}$	$\{3,4\}$	$\{4,5\}$	$\{5,6\}$	$\{6,7\}$	$\{7,8\}$	$\{8,9\}$.

Lemma 2. *After adding a vertex x, the coloring c has exactly two proper extensions with palette $\{1,\dots,9\}$. Their respective lists of colors on $x0,\dots,x7$ are*

$$(2,3,4,5,6,7,8,9) \quad \textit{and} \quad (9,2,3,4,5,6,7,8).$$

Proof. Write $f(v)$ for the color assigned to xv. Properness at v requires $f(v) \in M(v)$, and properness at x requires the eight values of f to be distinct.

If $f(0) = 2$, distinctness forces $f(1) = 3$, then $f(2) = 4$, and successively $f(3) = 5,\dots,f(7) = 9$. If $f(0) = 9$, distinctness instead forces $f(7) = 8$, then $f(6) = 7$, and successively $f(5) = 6,\dots,f(1) = 2$. These exhaust the two possibilities for $f(0)$. Each resulting assignment uses distinct colors and chooses a member of $M(v)$ at every vertex, so each is a proper extension. □

Proof of Theorem 1. Lemma 1 establishes properness and acyclicity of c. By Lemma 2, only two proper extensions need be considered.

In the first extension, $f(0) = 2$ and $f(2) = 4$. The cycle

$$x - 0 - 4 - 5 - 6 - 2 - x$$

has successive edge colors $2, 4, 2, 4, 2, 4$.

In the second extension, $f(1) = 2$ and $f(3) = 4$. The cycle

$$x - 1 - 7 - 3 - x$$

has successive edge colors $2, 4, 2, 4$. Neither extension is acyclic. □

References

[1] N. Alon, B. Sudakov, and A. Zaks, Acyclic edge colorings of graphs, *Journal of Graph Theory* **37** (2001), 157–167. doi:10.1002/jgt.1010.